A short proof of Lemma 68, after Deligne's tame case

(the generic Abel–Jacobi fiber is an affine space, and affine space has no prime-to-p abelian covers)

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Goal. The proof of Theorem 2 ("Theorem 2") reduces, in the ramified case $\mathfrak{m} > 0$, to the extension lemma whose proof in the thesis is currently a placeholder (`ProofOfReducedTheorem.tex`): a locally constant sheaf $\mathcal{F}^{[d]}$ on $U^{(d)}$ which is *tamely* ramified along the boundary divisor $H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$ extends to a locally constant sheaf on all of $\widetilde{C}^{(d)}_{\mathfrak{m}}$.

This note gives a proof modelled on the way Deligne's *tame* case of geometric class field theory is finished in ([Ten15], last section) and in ([Gui19], Section 5): there, after the unramified case, the tame case is deduced from two soft facts — the fibers of the open Abel–Jacobi map are affine spaces, and the abelianized fundamental group of affine space is pro-p, so a character of prime-to-p order is automatically constant on the fibers. The wild case cannot be reduced to the unramified one on the curve itself (no prime-to-p power of a wild character becomes unramified), which is why the thesis needs Takeuchi's compactification and the tameness theorem. But *at the boundary point* η_0 the same two soft facts apply verbatim and finish the proof: the prime-to-p inertia dies because the torsor is constant on the generic fiber of Φ_d, the p-primary inertia dies by tameness alone, and purity of the branch locus converts "unramified at η_0" into the extension. This realizes **Route 1** of the placeholder note: the kernel of $\pi_1^{ab}(U^{(d)}) \to \pi_1^{ab}(\mathrm{Pic}^d_{C,\mathfrak{m}})$ can meet our torsor only in its pro-p part.

All numbered references to results *not* proved in this note (Takeuchi's compactification, the degree condition, the tameness definitions, purity inputs, etc.) carry the numbering of the compiled thesis (`main.pdf`).

The extension lemma, after Deligne's tame case

We keep the notation of Section 3.2: C is a projective smooth geometrically connected curve over the perfect field k, $\mathfrak{m} = \sum n_i P_i > 0$ is a modulus with complement $U = C \setminus \mathfrak{m}$, and d is an integer satisfying (3.2). We write $Z_0 = Z_0(\mathfrak{m}, d) \subset C^{(d)}$ for the center of the blowup $\pi : X_{\mathfrak{m},d} \to C^{(d)}$, E_0 for its exceptional divisor with generic point η_0, $\widetilde{C}^{(d)}_{\mathfrak{m}} \subseteq X_{\mathfrak{m},d}$ for Takeuchi's compactification (Theorem 60), and $H = \widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)} = E_0 \times_{C^{(d)}} \widetilde{C}^{(d)}_{\mathfrak{m}}$ for the boundary divisor. Throughout, p denotes

the characteristic of k; if $p = 0$, statements of the form "prime to p" are to be read as no condition, and the arguments below only simplify. We set

$$N \;:=\; d - \deg \mathfrak{m} - g + 1 \;\geq\; 1,$$

the relative dimension of Φ_d (Section 3.1; positivity follows from (3.2)).

Lemma 1. *If $\mathcal{F}^{[d]}$ is a locally constant sheaf on $U^{(d)}$ which is tamely ramified along the boundary divisor $H = \tilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$, then $\mathcal{F}^{[d]}$ extends to a locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ on $\tilde{C}^{(d)}_{\mathfrak{m}}$.*

The proof has the same two-step skeleton as the other routes — by Zariski–Nagata purity it suffices to show the corresponding torsor is *unramified* at η_0 (Proposition 4), and tameness at η_0 is then upgraded to unramifiedness (Proposition 5) — but the second step is now soft. The point η_0 lies in the *generic fiber* of the compactified Abel–Jacobi morphism $\tilde{\Phi}_d$, with the same local ring; the corresponding generic fiber of the *open* morphism Φ_d is an honest affine space $\mathbb{A}^N_\kappa$ over the function field κ of $\mathrm{Pic}^d_{C,\mathfrak{m}}$ (Lemma 2). Since affine space has no nontrivial abelian covers of degree prime to p (Lemma 3 — five lines of Kummer theory), the prime-to-p part of our torsor is *constant* on that fiber, hence unramified at η_0; and the p-primary part of the inertia is killed by tameness alone, since a Galois extension of p-power order in residue characteristic p admits no nontrivial tame ramification. No valuation computation and no degree count is needed.

Geometry of the boundary and of the generic fiber

Let ξ denote the generic point of $\mathrm{Pic}^d_{C,\mathfrak{m}}$ and let $\kappa := \kappa(\xi)$ be its residue field, i.e. the function field of $\mathrm{Pic}^d_{C,\mathfrak{m}}$.

Lemma 2. *With notation as above:*

1. *$\widetilde{C}^{(d)}_{\mathfrak{m}}$ is an integral scheme, smooth over k; H is a nonempty open subscheme of E_0, irreducible of codimension 1 in $\widetilde{C}^{(d)}_{\mathfrak{m}}$ with generic point η_0; the point η_0 is the unique point of $\widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$ whose local ring has dimension 1, and $\mathcal{O} := \mathcal{O}_{\widetilde{C}^{(d)}_{\mathfrak{m}},\eta_0}$ is a discrete valuation ring with fraction field $K := k(C^{(d)})$.*

2. *$\tilde{\Phi}_d(\eta_0) = \xi$. The generic fiber $P_\xi := \tilde{\Phi}_d^{-1}(\xi)$ is isomorphic to $\mathbb{P}^N_\kappa$, its function field is K, and the local ring of P_ξ at η_0* equals *$\mathcal{O}$. In particular $\kappa \subseteq \mathcal{O}$.*

3. *$A_\xi := P_\xi \cap U^{(d)} = \Phi_d^{-1}(\xi)$ is a dense open subscheme of P_ξ with closed complement $P_\xi \cap H$, its generic point is* $\mathrm{Spec}(K)$, *and*

$$A_\xi \;\cong\; \mathbb{A}^N_\kappa$$

as a κ-scheme.

Proof. (1) $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is an open subscheme of the blowup $X_{\mathfrak{m},d}$ of the integral scheme $C^{(d)}$, hence integral, with function field K; it is a projective space bundle over the smooth k-scheme $\mathrm{Pic}^d_{C,\mathfrak{m}}$ (Theorem 60), hence smooth over k. The base change $H = E_0 \times_{X_{\mathfrak{m},d}} \widetilde{C}^{(d)}_{\mathfrak{m}} \hookrightarrow E_0$ of the open immersion $\widetilde{C}^{(d)}_{\mathfrak{m}} \hookrightarrow X_{\mathfrak{m},d}$ is an open immersion, and $H \neq \emptyset$: otherwise $\Phi_d = \tilde{\Phi}_d$ would be proper, while its fibers are affine spaces of dimension $N \geq 1$ (Section 3.1). A nonempty open subscheme of the irreducible E_0 is irreducible with the same generic point η_0; every other point of H is a proper specialization of η_0, so has local ring of dimension ≥ 2. Finally $\mathcal{O}$ is a one-dimensional regular local ring, i.e. a DVR, with fraction field $k(\widetilde{C}^{(d)}_{\mathfrak{m}}) = K$.

(2) Every fiber of $\tilde{\Phi}_d$ meets H: a fiber contained in $U^{(d)}$ would be a proper scheme equal to a fiber of Φ_d, which is a positive-dimensional affine space over the separable closure — not proper. Hence $\tilde{\Phi}_d(H) = \mathrm{Pic}^d_{C,\mathfrak{m}}$ ($\tilde{\Phi}_d$ is proper, so the image of the closed set H is closed and, as just shown, it meets every point). Since $H = \overline{\{\eta_0\}} \cap \widetilde{C}^{(d)}_{\mathfrak{m}}$,

$$\mathrm{Pic}^d_{C,\mathfrak{m}} = \tilde{\Phi}_d(H) \subseteq \overline{\{\tilde{\Phi}_d(\eta_0)\}},$$

which forces $\tilde{\Phi}_d(\eta_0) = \xi$. Next, since ξ is the generic point of the integral scheme $\mathrm{Pic}^d_{C,\mathfrak{m}}$, its local ring is the field κ, so

$$P_\xi = \widetilde{C}^{(d)}_{\mathfrak{m}} \times_{\mathrm{Pic}^d_{C,\mathfrak{m}}} \operatorname{Spec} \mathcal{O}_{\mathrm{Pic}^d_{C,\mathfrak{m}},\xi} = \varprojlim_{V \ni \xi \text{ open}} \tilde{\Phi}_d^{-1}(V)$$

is a filtered limit of *open* subschemes of $\widetilde{C}^{(d)}_{\mathfrak{m}}$ along the identity maps; consequently, for every point $y \in P_\xi$ the local rings agree, $\mathcal{O}_{P_\xi,y} = \mathcal{O}_{\widetilde{C}^{(d)}_{\mathfrak{m}},y}$. Applying this to $y = \eta_0$ gives $\mathcal{O}_{P_\xi,\eta_0} = \mathcal{O}$, and in particular $\kappa = \mathcal{O}_{\mathrm{Pic}^d_{C,\mathfrak{m}},\xi} \subseteq \mathcal{O}$. By Theorem 60, $\widetilde{C}^{(d)}_{\mathfrak{m}} = \mathbb{P}(\mathcal{E})$ for a locally free sheaf $\mathcal{E}$ on $\mathrm{Pic}^d_{C,\mathfrak{m}}$, so $P_\xi = \mathbb{P}(\mathcal{E}_\xi) \cong \mathbb{P}^N_\kappa$ (the fiber dimension is N because P_ξ contains the fiber of Φ_d, of dimension N, as a dense open subset — see (3)). The generic point of $\widetilde{C}^{(d)}_{\mathfrak{m}}$ maps to ξ (dominance) and lies in P_ξ as its generic point, so $\kappa(P_\xi) = K$.

(3) $P_\xi \cap U^{(d)} = P_\xi \setminus H$ holds because $\widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)} = H$ (Theorem 60), and this set equals $\Phi_d^{-1}(\xi)$ because $\tilde{\Phi}_d$ extends Φ_d. It is open in the irreducible P_ξ and nonempty (Φ_d is surjective, Section 3.1), hence dense with the same generic point $\operatorname{Spec} K$.

For the last isomorphism, recall the modular description of the fibers of Φ_d ([Gui19, Section 4, proof of Theorem 4.14]; it is also the description recalled in Section 3.1). Since $\mathfrak{m}_\kappa \to \operatorname{Spec} \kappa$ is finite flat and surjective, the rigidified Picard presheaf is already a sheaf, so the κ-point ξ of $\mathrm{Pic}^d_{C,\mathfrak{m}}$ is represented by an actual pair $(\mathcal{L}, \psi)$ on C_κ. For a κ-scheme T, a T-point of $\Phi_d^{-1}(\xi)$ is a divisor $D \in U^{(d)}(T)$ together with the (unique, since rigidified pairs have no nontrivial automorphisms) isomorphism $(\mathcal{O}_C(D), \psi_D) \cong (\mathcal{L}, \psi)_T$; sending D to the image $s \in \Gamma(C_T, \mathcal{L}_T)$ of the canonical section 1_D identifies $\Phi_d^{-1}(\xi)(T)$ with

$$V_\psi(T) := \{\, s \in \Gamma(C_T, \mathcal{L}_T) \;:\; s|_{\mathfrak{m}_T} = \psi_T(1) \,\}$$

(the condition $s|_{\mathfrak{m}_T} = \psi_T(1)$ pins down the scaling, and such an s is fiberwise nonzero because it is invertible along $\mathfrak{m}$, so $D = V(s)$ is a relative effective Cartier divisor of degree d prime to $\mathfrak{m}$). By the degree condition, on each fiber $H^1(C, \mathcal{L}(-\mathfrak{m})) = 0$, so (cohomology and base change, as recalled in Section 3.2) the restriction map of vector groups $\mathrm{pr}_*\mathcal{L} \to \mathrm{pr}_*(\mathcal{L}|_{\mathfrak{m}})$ over κ is surjective with kernel the vector group of $W := H^0(C_\kappa, \mathcal{L}(-\mathfrak{m}))$, of dimension N (Riemann–Roch). Hence $A_\xi \cong V_\psi$ is a torsor under the vector group $\mathbb{W} \cong \mathbb{G}_a^N$ over κ. Every such torsor is trivial, since $H^1_{et}(\operatorname{Spec} \kappa, \mathbb{G}_a^N) = H^1(\mathrm{Gal}(\kappa^{sep}/\kappa), (\kappa^{sep})^N) = 0$ (additive Hilbert 90). Therefore $A_\xi \cong \mathbb{W} \cong \mathbb{A}^N_\kappa$. □

Affine space has no prime-to-p abelian covers

The following is the elementary engine of Deligne's tame case, as used in ([Ten15], last section: "$\pi_1(\mathbb{A}^n)^{ab}$ is a pro-p group") and in ([Gui19], Section 5: triviality of the tame fundamental group of the affine line).

Lemma 3. *Let F be a separably closed field and let G' be a finite abelian group whose order is invertible in F. Then every G'-torsor on $(\mathbb{A}^N_F)_{et}$ is trivial. Equivalently, every quotient of $\pi_1^{et,ab}(\mathbb{A}^N_F)$ of order invertible in F vanishes (for $\operatorname{char} F = p > 0$: $\pi_1^{et,ab}(\mathbb{A}^N_F)$ is pro-p).*

Proof. Writing $G' \cong \prod_i \mathbb{Z}/n_i\mathbb{Z}$ and decomposing the torsor accordingly (Proposition 15), we may assume $G' = \mathbb{Z}/n\mathbb{Z}$ with n invertible in F. Since F is separably closed and $x^n - 1$ is separable, F contains all n-th roots of unity, so $\mathbb{Z}/n\mathbb{Z} \cong \mu_n$ as étale sheaves on $\mathbb{A}^N_F$, and it suffices to show $H^1_{et}(\mathbb{A}^N_F, \mu_n) = 0$ (étale H^1 with coefficients in a finite abelian constant group classifies torsors under it, cf. Corollary 14). The Kummer sequence $1 \to \mu_n \to \mathbb{G}_m \xrightarrow{n} \mathbb{G}_m \to 1$, exact on the étale site because n is invertible, gives

$$\Gamma(\mathbb{A}^N_F, \mathcal{O})^\times \xrightarrow{n} \Gamma(\mathbb{A}^N_F, \mathcal{O})^\times \longrightarrow H^1_{et}(\mathbb{A}^N_F, \mu_n) \longrightarrow \operatorname{Pic}(\mathbb{A}^N_F)[n].$$

Now $\Gamma(\mathbb{A}^N_F, \mathcal{O})^\times = F[x_1, \dots, x_N]^\times = F^\times$, which is n-divisible ($x^n - a$ is separable for $a \neq 0$, and F is separably closed), and $\operatorname{Pic}(\mathbb{A}^N_F) = 0$ because $F[x_1, \dots, x_N]$ is a unique factorization domain. Hence $H^1_{et}(\mathbb{A}^N_F, \mu_n) = 0$. □

Step 1: purity, and extending unramified torsors

This step is identical to the one in the other two routes; we reproduce it for completeness.

Proposition 4. *Let G be a finite abelian group, let $\rho : \pi_1^{et}(U^{(d)}, \bar{u}) \to G$ be a continuous homomorphism and let $\mathcal{Q}$ be the corresponding G-torsor on $U^{(d)}_{ét}$ (Corollary 14). If $\mathcal{Q}$ is unramified over $\widetilde{C}^{(d)}_{\mathfrak{m}}$ at (H, η_0) in the sense of Definition 40, then ρ factors, necessarily uniquely, through the canonical surjection $\iota_* : \pi_1^{et}(U^{(d)}, \bar{u}) \twoheadrightarrow \pi_1^{et}(\widetilde{C}^{(d)}_{\mathfrak{m}}, \bar{u})$, where $\iota : U^{(d)} \hookrightarrow \widetilde{C}^{(d)}_{\mathfrak{m}}$ is the open immersion. Consequently, $\mathcal{Q}$ extends to a G-torsor on $(\widetilde{C}^{(d)}_{\mathfrak{m}})_{ét}$, uniquely up to isomorphism.*

Proof. **Surjectivity of ι_*.** For every connected finite étale cover $T \to \widetilde{C}^{(d)}_{\mathfrak{m}}$, the scheme T is regular (étale over the regular $\widetilde{C}^{(d)}_{\mathfrak{m}}$), hence irreducible, and $T \times_{\widetilde{C}^{(d)}_{\mathfrak{m}}} U^{(d)}$ is a nonempty open subscheme of T, hence connected. If the image of ι_* were contained in a proper open subgroup $M \leq \pi_1^{et}(\widetilde{C}^{(d)}_{\mathfrak{m}}, \bar{u})$, the connected cover corresponding to M would pull back to a disconnected cover of $U^{(d)}$ — a contradiction.

Normalization. Let $G' = \rho(\pi_1^{et}(U^{(d)}, \bar{u}))$ and let $V \to U^{(d)}$ be the connected Galois cover with group G' corresponding to $\ker \rho$ (Theorem 13); as a $U^{(d)}$-scheme, $\mathcal{Q}$ is a disjoint union of $[G : G']$ copies of V, so V is unramified over $\widetilde{C}^{(d)}_{\mathfrak{m}}$ at (H, η_0) as well. V is integral and $K(V)/K$ is finite separable. Let $\nu : W \to \widetilde{C}^{(d)}_{\mathfrak{m}}$ be the normalization of $\widetilde{C}^{(d)}_{\mathfrak{m}}$ in $K(V)$; since $\widetilde{C}^{(d)}_{\mathfrak{m}}$ is Nagata ([Stacks, Tag 0335]), ν is finite, W is integral and normal, and $\nu^{-1}(U^{(d)}) \cong V$ by uniqueness of normalization.

ν is étale, by purity of the branch locus. We apply [Stacks, Tag 0BMB] at each point $w \in W$ over $y = \nu(w)$: $\mathcal{O}_{W,w}$ is normal, $\mathcal{O}_{\widetilde{C}^{(d)}_{\mathfrak{m}},y}$ is regular, ν is finite hence quasi-finite, and $\dim(\mathcal{O}_{W,w}) = \dim(\mathcal{O}_{\widetilde{C}^{(d)}_{\mathfrak{m}},y})$ by the dimension formula for the finite dominant morphism ν of integral schemes of finite type over k. For the codimension-one condition, let $w' \rightsquigarrow w$ with $\dim(\mathcal{O}_{W,w'}) = 1$ and $y' = \nu(w')$, so $\dim(\mathcal{O}_{\widetilde{C}^{(d)}_{\mathfrak{m}},y'}) = 1$. If $y' \in U^{(d)}$, then ν is étale at w' because $\nu^{-1}(U^{(d)}) = V$ is étale over $U^{(d)}$. Otherwise $y' = \eta_0$ by Lemma 2(1), and $\mathcal{O}_{W,w'}$ is a localization of the integral closure of the discrete valuation ring $\mathcal{O}$ in $K(V)$; the hypothesis that V is unramified at (H, η_0) says precisely (Definition 37) that the ramification index there is 1 with separable residue extension, i.e. that ν is unramified at w'. Thus [Stacks, Tag 0BMB] applies and ν is finite étale.

Factorization. As $V = W \times_{\widetilde{C}^{(d)}_{\mathfrak{m}}} U^{(d)}$, the action of $\pi_1^{et}(U^{(d)}, \bar{u})$ on the fiber $V_{\bar{u}} = W_{\bar{u}}$ is the restriction along ι_* of the $\pi_1^{et}(\widetilde{C}^{(d)}_{\mathfrak{m}}, \bar{u})$-action; the kernel of the former is $\ker\rho$, so $\ker(\iota_*) \subseteq \ker\rho$ and ρ factors as $\rho = \tilde{\rho} \circ \iota_*$, uniquely by surjectivity of ι_*. The G-torsor $\tilde{\mathcal{Q}}$ corresponding to $\tilde{\rho}$ restricts to $\mathcal{Q}$ on $U^{(d)}$, and any two extensions have equal characters, hence are isomorphic. □

Step 2: at the boundary, tame implies unramified

Proposition 5. *Let G be a finite abelian group and let $\mathcal{Q}$ be a G-torsor on $U^{(d)}_{\acute{e}t}$. If $\mathcal{Q}$ is tamely ramified over $\widetilde{C}^{(d)}_{\mathfrak{m}}$ at (H, η_0), then $\mathcal{Q}$ is unramified at (H, η_0).*

Note that no bound on the ramification of $\mathcal{Q}$ along $\mathfrak{m}$ is assumed, and that tameness is used only to kill the p-primary part of the inertia: the prime-to-p part dies unconditionally, by the geometry of the generic fiber.

Proof. Let $\widehat{K} = \widehat{K}_{\eta_0}$ be the fraction field of the completion of the discrete valuation ring $\mathcal{O}$ (Lemma 2(1)), and let

$$\rho : G_{\widehat{K}} \longrightarrow G$$

be the continuous homomorphism classifying the restriction of $\mathcal{Q}$ to $\mathrm{Spec}(\widehat{K})$. Let $\widehat{K}^{ur} \subseteq \widehat{K}^{sep}$ be the maximal unramified extension and $I := \mathrm{Gal}(\widehat{K}^{sep}/\widehat{K}^{ur})$ the inertia subgroup; the assertion to be proved is $\rho(I) = \{1\}$ (Definition 37).

Tameness confines the inertia to the prime-to-p part. Write $G = G_p \times G'$ with G_p a p-group and $|G'|$ prime to p. The fixed field of $\ker\rho$ is a finite Galois extension $F/\widehat{K}$ with group $\rho(G_{\widehat{K}}) \subseteq G$ and inertia subgroup $\rho(I)$. By hypothesis this extension is tame: its residue extension is separable and its ramification index e is prime to p. For such extensions $|\rho(I)| = e$ ([Ser79, Ch. IV]), so $\rho(I)$ is a finite group of order prime to p, whence $\rho(I) \subseteq \{1\} \times G'$. Letting $\rho' := \mathrm{pr}_{G'} \circ \rho$, which classifies the quotient G'-torsor $\mathcal{Q}' := \mathcal{Q}/G_p$ (Proposition 15), it therefore suffices to prove $\rho'(I) = \{1\}$.

The prime-to-p part is constant on the generic fiber. By Lemma 2, $A_\xi \cong \mathbb{A}^N_\kappa$ is a subscheme of $U^{(d)}$ whose generic point is $\mathrm{Spec}(K)$, and $\kappa \subseteq \mathcal{O} \subseteq \widehat{K}$. Consider the restriction $\mathcal{Q}'|_{A_\xi}$, a G'-torsor on $\mathbb{A}^N_\kappa$. Its pullback to $\mathbb{A}^N_{\kappa^{sep}}$ is trivial by Lemma 3 ($|G'|$ is invertible in κ, since it is prime to p). A trivializing section is a section of the finite étale scheme $\mathcal{Q}'|_{A_\xi}$ over $\mathbb{A}^N_{\kappa^{sep}} = \varprojlim_{\kappa'} \mathbb{A}^N_{\kappa'}$, the limit running over the finite separable subextensions κ'/κ of κ^{sep}; as $\mathcal{Q}'|_{A_\xi}$ is of finite presentation, the section is defined at some finite level. Hence there is a finite separable extension κ'/κ such that

$$\mathcal{Q}'|_{A_\xi \times_\kappa \kappa'} \text{ is the trivial } G'\text{-torsor.}$$

Constant covers are unramified. Let $\widehat{\mathcal{O}}$ be the completion of $\mathcal{O}$, so $\widehat{K} = \mathrm{Frac}(\widehat{\mathcal{O}})$ and $\kappa \subseteq \mathcal{O} \subseteq \widehat{\mathcal{O}}$. Since κ'/κ is finite separable, the base change $\widehat{\mathcal{O}} \otimes_\kappa \kappa'$ is finite étale over the complete discrete valuation ring $\widehat{\mathcal{O}}$; hence it is a finite product of complete discrete valuation rings, each of which is unramified over $\widehat{\mathcal{O}}$ (étale means exactly: the maximal ideal of $\widehat{\mathcal{O}}$ generates the maximal ideal, i.e. the ramification index is 1, and the residue extension is separable). Inverting a uniformizer, $\widehat{K} \otimes_\kappa \kappa' = \prod_j \widehat{K}_j$ is a finite product of finite separable field extensions of $\widehat{K}$, each *unramified* over $\widehat{K}$. Fix one factor $\widehat{K}_1$ and embed it in $\widehat{K}^{sep}$; then $\widehat{K}_1 \subseteq \widehat{K}^{ur}$, so

$$I \;=\; \mathrm{Gal}(\widehat{K}^{sep}/\widehat{K}^{ur}) \;\subseteq\; \mathrm{Gal}(\widehat{K}^{sep}/\widehat{K}_1) \;=\; G_{\widehat{K}_1}.$$

Now $\mathrm{Spec}(\widehat{K}_1) \to \mathrm{Spec}(\widehat{K} \otimes_\kappa \kappa') \to \mathrm{Spec}(K \otimes_\kappa \kappa') \to A_\xi \times_\kappa \kappa'$ (the last map is the inclusion of a generic point, Lemma 2(3)), so the restriction of ρ' to $G_{\widehat{K}_1}$ classifies the pullback of the *trivial* torsor $\mathcal{Q}'|_{A_\xi \times_\kappa \kappa'}$, and is therefore the trivial homomorphism:

$$\rho'\Big(G_{\widehat{K}_1}\Big) = \{1\}, \qquad \text{hence} \qquad \rho'(I) \subseteq \rho'\Big(G_{\widehat{K}_1}\Big) = \{1\}.$$

Combining with the first step, $\rho(I) \subseteq \{1\} \times G'$ has trivial image under $\mathrm{pr}_{G'}$, so $\rho(I) = \{1\}$, i.e. $\mathcal{Q}$ is unramified at (H, η_0). □

Conclusion of the proof

Proof of Lemma 1. Let $G = \Lambda^\times$ and let $\mathcal{P}^{[d]}$ be the G-torsor on $U^{(d)}_{\text{ét}}$ corresponding to $\mathcal{F}^{[d]}$ under Proposition 9, with character $\rho : \pi_1^{et}(U^{(d)}, \bar{u}) \to G$. By Lemma 2(1), the only codimension-one point of $\widetilde{C}^{(d)}_{\mathfrak{m}} \setminus U^{(d)}$ is η_0, so the hypothesis that $\mathcal{F}^{[d]}$ is tamely ramified along H means precisely that $\mathcal{P}^{[d]}$ is tamely ramified over $\widetilde{C}^{(d)}_{\mathfrak{m}}$ at (H, η_0) (Definition 40). By Proposition 5, $\mathcal{P}^{[d]}$ is unramified at (H, η_0), and by Proposition 4 the character ρ factors uniquely through $\pi_1^{et}(\widetilde{C}^{(d)}_{\mathfrak{m}}, \bar{u})$. The locally constant sheaf $\tilde{\mathcal{F}}^{[d]}$ of free rank-one Λ-modules corresponding to the factored character (Corollary 14 together with Proposition 9) restricts to $\mathcal{F}^{[d]}$ on $U^{(d)}$. □

Remark. As in the other routes, the proof also gives uniqueness of the extension, and full faithfulness of restriction from locally constant sheaves on $\widetilde{C}^{(d)}_{\mathfrak{m}}$ to locally constant sheaves on $U^{(d)}$, since $\iota_* : \pi_1^{et}(U^{(d)}) \to \pi_1^{et}(\widetilde{C}^{(d)}_{\mathfrak{m}})$ is surjective (Proposition 4). This is the source of the uniqueness assertions in Theorem 70 and Theorem 2.

Remark (Relation to Deligne's tame case, and to Routes 1 and 2). In Deligne's tame case (modulus $\mathfrak{m}$ reduced; [Ten15], last section) the two soft inputs above are used directly on the open Abel–Jacobi fibration: a tame character becomes unramified on the curve after raising to a prime-to-p power, the unramified case descends it, and pro-p-ness of π_1^{ab} of the affine-space fibers transfers the conclusion back to the character itself. With wild ramification this reduction is impossible *on the curve* — no prime-to-p power of a wildly ramified character is unramified — which is why the thesis passes through Takeuchi's compactification and the tameness theorem. The proof above is the same mechanism executed once, at the single boundary point η_0: tameness (the thesis's hard input) kills the p-part of the inertia, and the affine-space geometry of the generic Abel–Jacobi fiber kills the prime-to-p part. In the language of the placeholder note, this is **Route 1**: the kernel of $\pi_1^{ab}(U^{(d)}) \to \pi_1^{ab}(\mathrm{Pic}^d_{C,\mathfrak{m}})$ meets our character only in its pro-p part. Route 2 (`lemma_88_route_2/`) and the direct valuation computation (`extension_lemma_completion/`) prove the same statement; the present route replaces their Kummer-valuation and degree-count arguments by the triviality statement Lemma 3.

The global geometry is indispensable: for a proper submodulus $\mathfrak{n} \subsetneq \mathfrak{m}$ the analogous statement at $\eta_{\mathfrak{n}}$ is *false* (the computations of Chapter 2 produce genuinely, tamely, ramified symmetric powers there), and correspondingly the exceptional divisor of the blowup at $\mathfrak{n}$ does not sit inside a fibration with affine-space fibers over a base of the descent.

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